Lunar South Pole Mission Plan

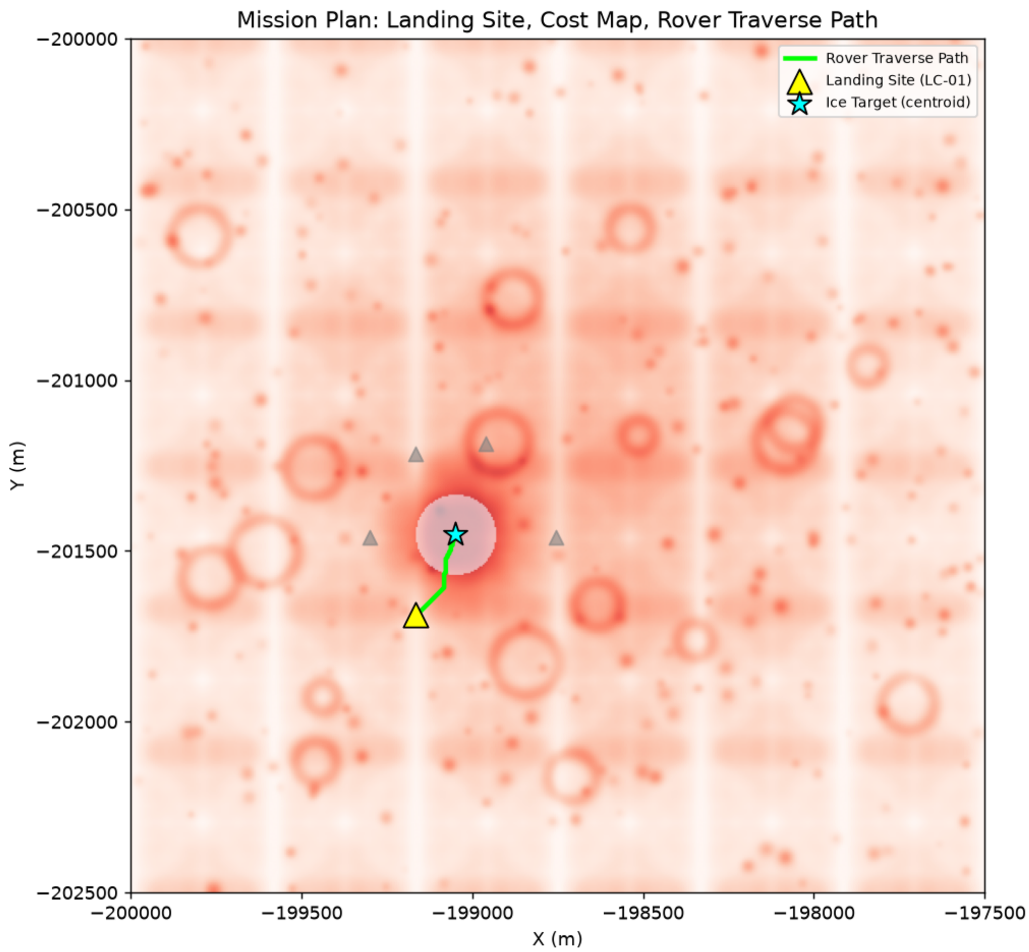
Rover Traverse Planning & Landing Site Selection

1. Landing Site Candidate Comparison

Candidate	Safety	Access	Science	Final Score
LC-01	1.000	0.770	0.764	0.884
LC-02	0.863	0.770	0.764	0.815
LC-03	0.659	1.000	0.764	0.782
LC-04	0.542	0.000	0.764	0.424
LC-05	0.000	0.381	0.764	0.267

Selected site: **LC-01** (Final Score = 0.884). Scoring weights: Safety 50%, Accessibility 30%, Science Value 20%, reflecting that mission/crew safety is prioritized while still favoring proximity to high-confidence ice.

2. Mission Map



3. Traverse Summary

Metric	Value
Total traverse distance	282.6 m
Estimated travel time	1.57 hours
Avg. shadow/hazard exposure along path	37.2%
Estimated energy consumption	53.9 Wh
Assumed rover speed	180 m/h
Path planning algorithm	A* (8-connected grid, cost-weighted)

4. Methodology & Limitations

Path planning uses an 8-connected A* search over a cost-weighted grid derived from the HazardMap (shadow depth, crater rims, boulder density, surface roughness). No DEM was available for this analysis; consequently, energy estimates do not account for terrain slope or elevation change and rely on a simplified power model scaled by hazard/shadow exposure along the path. Dijkstra and Hybrid A* were considered as alternatives: Dijkstra guarantees a global optimum without a heuristic but is slower, while Hybrid A* would additionally respect rover turning-radius constraints and is recommended as a production-grade upgrade once kinematic rover parameters are available.